

Low-fidelity Prototyping

Creating tangible representations of ideas

ACADEMIC RESOURCES:

Sefelin, R., Tscheligi, M., & Giller, V. (2003). Paper prototyping-what is it good for?: a comparison of paper-and computer-based low-fidelity prototyping. In CHI'03 extended abstracts on Human factors in computing systems (pp. 778-779). ACM.

Svanaes, D., & Seland, G. (2004). Putting the users center stage: role playing and low-fi prototyping enable end users to design mobile systems. In Proceedings of the SIGCHI conference on Human factors in computing systems (pp. 479-486). ACM.

Warfel, T. Z. (2009). Prototyping: a practitioner's guide. New York, NY: Rosenfeld Media.

A prototype is a representation of an envisioned product. Low-fidelity prototyping allows the quick exploration of ideas early in a design process. They can be used to reflect on a design, to discuss design solutions within a team and to get feedback from prospective users through usability testing (p.160).

A prototype should represent the actual scale of the final product. This allows people to experience what the interaction with the final product would feel like, whether it is easy to use and whether there are aspects that need to be improved. They are typically hand-sketched, use various materials and do not represent the final visual design. Jeff Hawkins, the founder of Palm Computing, carried around a block of wood the size of a pocket computer, which later led to the invention of the hugely successful PalmPilot. To determine what features to include and how to design them, he

would take it to meetings and use a pen to interact with the prototype as if he was setting up calendar appointments.

Low-fidelity prototypes are different to mock-ups (p.108) as they do not represent the final visual design. They are also different from wireframes (p.174) as they use tangible materials to explore how people could interact with the final product or service. As such, low-fidelity prototypes often take the form of a 'horizontal' prototype, which is a prototype that represents the surface of a product interface but not any of the underlying technology. In comparison, 'vertical' prototypes only represent a small aspect of the product interface (for example, only the login page) but are fully implemented and functional – which can be useful to test technical aspects, such as loading times.

EXERCISE

YOU WILL NEED

Pen, marker, paper, tracing paper, scissors, sticky tape, Blu-tack



In this exercise, you will design a low-fidelity prototype that addresses a specific brief. Focus on your own design problem, or choose a design brief (p.180). If you have findings from user research (e.g. interviews, p.92), be sure to consider these in your design. See p.184 for an example of a low-fidelity prototype.

1

Sketch the parts of your product that you want to be **represented in the prototype**. Choose a specific task and sketch all the steps required to fulfil this task.

E.g. adding a person to your friends in a social network app
[15 minutes]

2

Create wireframes (p.174) of your product using pen and paper. These are an iteration of the sketches produced in the first step and show the key elements on each screen.

[30 minutes]

3

Create a physical prototype based on the previous step using any suitable materials.

E.g. for the social network app example, use the smartphone drawing templates provided on the companion website to create a paper version of each screen. Add dynamic features to allow for user interaction. This can be achieved either by swapping out a screen or by adding layers in the form of sticky notes. E.g. in a social network app, pressing the 'Find' button will bring up a new screen that lists people matching the name that was 'typed' into the search field.
[30 minutes]

4

Add real content by writing it directly into your prototype or on additional materials that can be overlaid, such as sticky notes.

E.g. in a screen showing search results in a social network app, add names that are made up but sound real. This will help to give the prototype a more realistic feel.
[15 minutes]

5

You can also **prototype other physical aspects**. For example, you could create a physical representation of a virtual reality headset with minimal materials.

E.g. using a cardboard VR headset

